

1. The table shows a list of the twenty largest springs in the state of Florida.

Name	Water District	Average Discharge (cubic feet/second)	Name	Water District	Average Discharge (cubic feet/second)
Spring Creek Springs	Northwest Florida	2,000	Homosassa Springs	Southwest Florida	192
Crystal River Springs	Southwest Florida	878	Blue Springs	Northwest Florida	189
Silver Springs	St. Johns River	799	Manatee Spring	Suwannee River	178
Rainbow Springs	St. Johns River	711	Weeki Wachee Springs	Southwest Florida	174
Alapaha Rise	Suwannee River	608	River Sink Spring	Northwest Florida	164
St. Marks Spring	Northwest Florida	517	Gainer Springs	Northwest Florida	159
Wakulla Springs	Northwest Florida	391	Blue Spring	St. Johns River	158
Wacissa Springs	Northwest Florida	288	Troy Spring	Suwannee River	152
Ichetucknee Springs	Suwannee River	360	Kini Springs	Northwest Florida	150
Holton Spring	Suwannee River	243	Hornsby Spring	Suwannee River	148

- a. Select all of the displays that can be used to display the data set of the water districts.

☐ Box plot

☒ Bar graph

☐ Histogram

☐ Line graph

☐ Scatter plot

☒ Circle graph

☐ Stem-and-leaf

☒ Frequency table

☐ Line plot (Dot plot)

☒ Relative Frequency table

- b. Complete the statement that justify your choices.

Because the data set of the water districts is

☐ numerical

☒ categorical

and

☐ bivariate,
☒ univariate,

the displays chosen in **part a** can be used.

- c. Select all of the displays that can be used to display the data set of the average discharge in cubic feet per second.

☒ Box plot

☐ Bar graph

☒ Histogram

☐ Line graph

☐ Scatter plot

☐ Circle graph

☒ Stem-and-leaf

☐ Frequency table

☒ Line plot (Dot plot)

☐ Relative Frequency table

- d. Complete the statement that justify your choices.

Because the data set of the water districts is

☒ numerical

☐ categorical

and

☐ bivariate,

☒ univariate,

the displays chosen in **part c** can be used.

2. A research survey of 2,006 public and private school students in the United States (grades 10–12) between April 12 and June 12, 2016 asked students if they agreed with the statement, “If I make a mistake, I try to figure out where I went wrong.” The survey found that 86% students agreed with the statement. The margin of error for the survey is $\pm 3.7\%$.

- a. Determine the range of surveyed students that agreed with the statement.

**Range of Surveyed
Students**

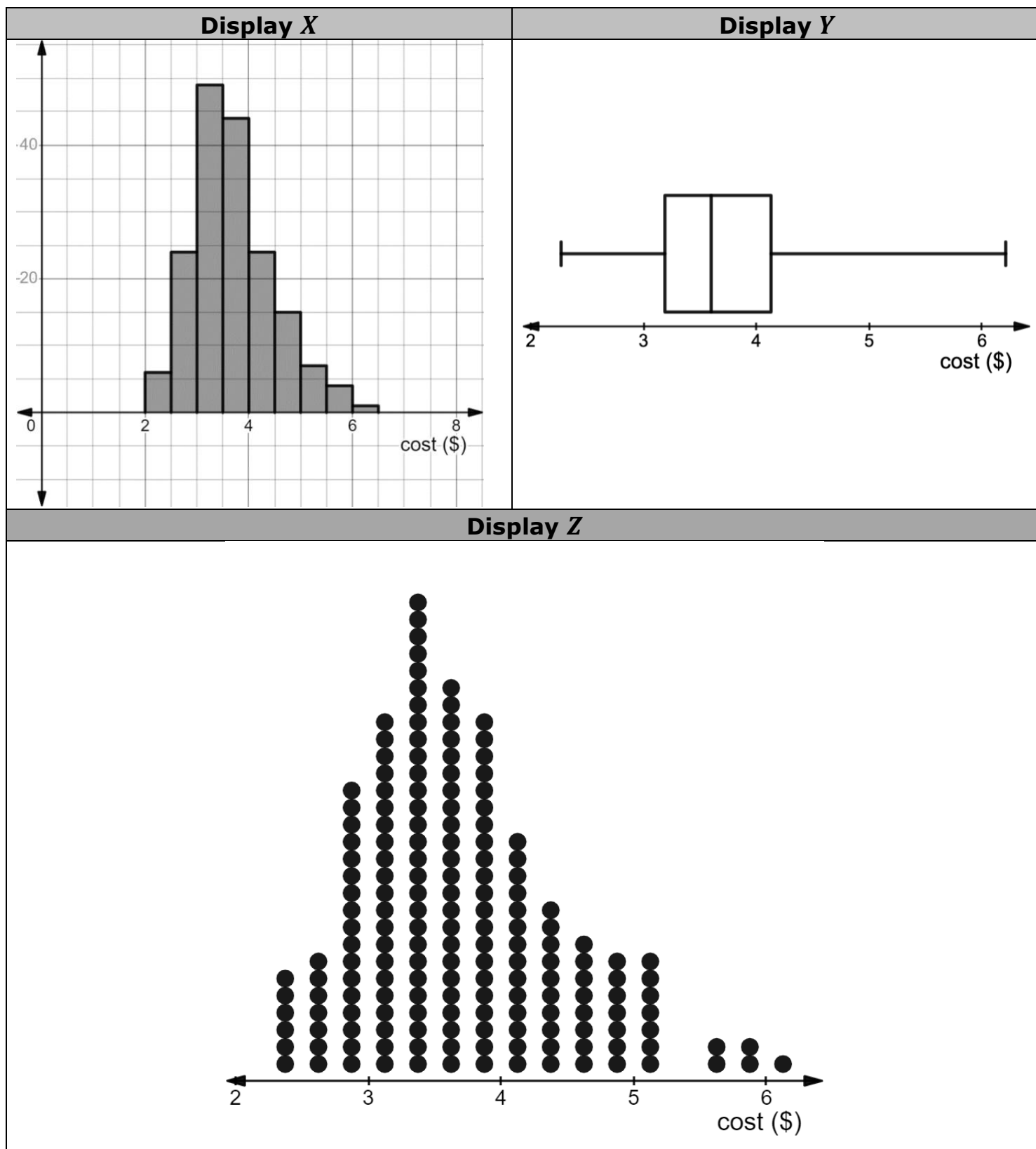
1651 to 1799 students

- b. In 2022, there were 47,851 students in grades 10-12 in Orange County Public Schools (OCPS). Determine the range of grade 10-12 students in OCPS who would likely agree with the statement, “If I make a mistake, I try to figure out where I went wrong.”

**Range of Students in Orange
County**

39,381 to 42,923 students

3. The daily cost of a healthy diet is the lowest-cost set of foods available that would meet requirements in dietary guidelines from governments and public health agencies around the world. The mean cost is \$3.70, the median cost is \$3.60, the minimum cost is \$2.27, the maximum cost is \$6.21, the first quartile is \$3.19, the third quartile is \$4.13, and the standard deviation is \$0.75. The data for 174 countries is displayed using three different types of displays.



Complete the statements.

a. The measure of center that best summarizes the cost of a healthy diet is the

- ☐ mean
- ☒ median

because display

- ☐ X
- ☒ Y
- ☐ Z

shows that the

- ☒ mean
- ☐ median

would be influenced by

- ☒ outliers.
- ☐ the range.
- ☐ the standard deviation.
- ☐ the interquartile range.

b. Displays

- ☐ X and Y
- ☐ X and Z
- ☐ Y and Z
- ☒ X , Y , and Z

can be used to show that the shape of the data is

- ☒ skewed
- ☐ bell-shaped

which indicates that the

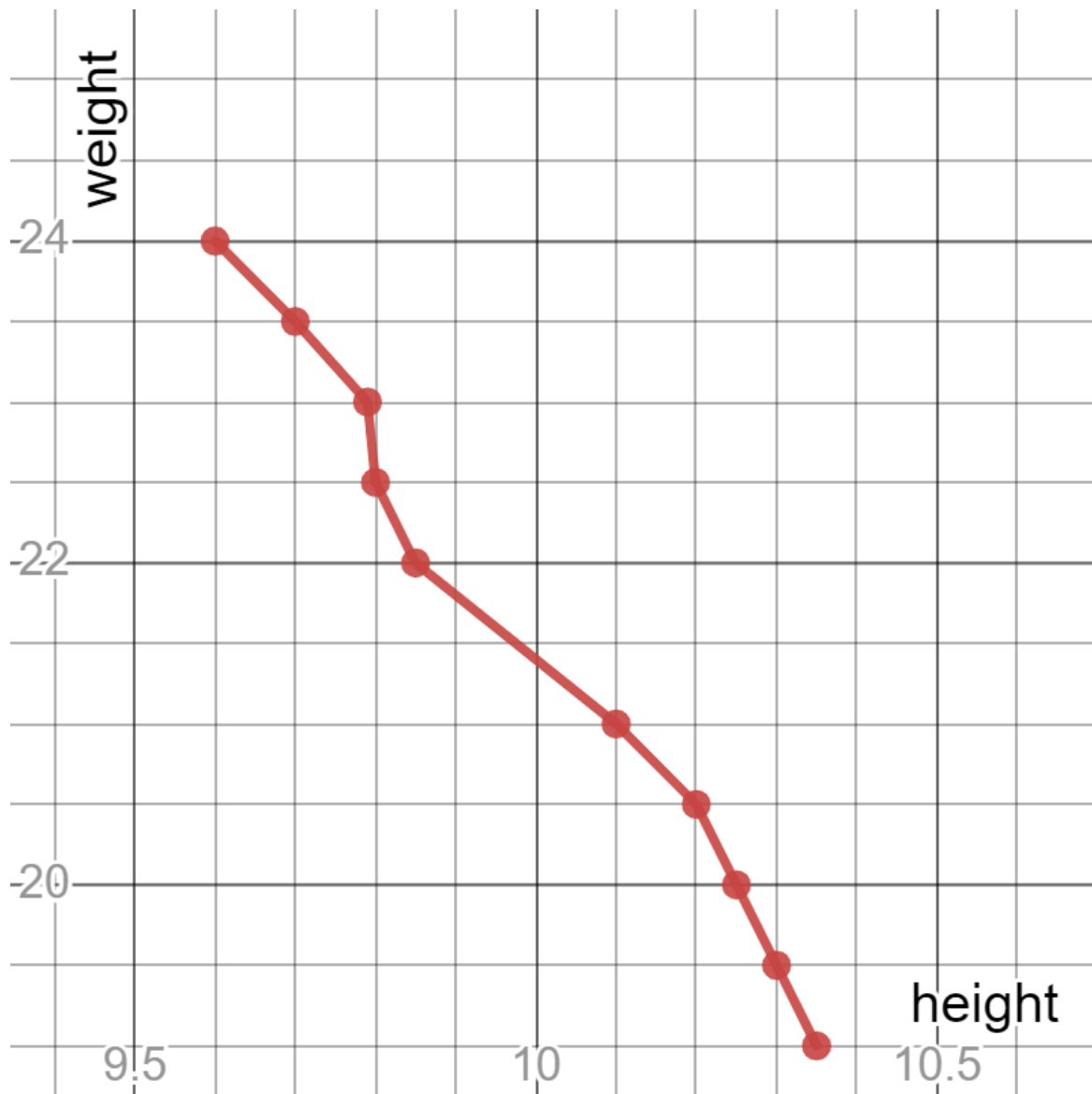
- ☐ range
- ☐ outliers
- ☐ standard deviation
- ☒ interquartile range

would be the best measure of variability.

4. The table shows the data of height, in inches (in.), a BMX bike can jump and the weight, in pounds (lbs), of the bike.

HEIGHT (in.)	WEIGHT (lbs)
10.35	19.0
10.30	19.5
10.25	20.0
10.20	20.5
10.10	21.0
9.85	22.0
9.80	22.5
9.79	23.0
9.70	23.5
9.60	24.0

Use the grid to create a line graph of the data.



5. A controlled study of the effects of cell phone usage of students aged 12–15 years old living in Wuhu City, China was performed during September and December 2014. The study found addiction to smartphones and poor sleep quality were correlated with a correlation coefficient of 0.906.

a. Complete the statement.

The value of the correlation coefficient shows that

- ☐ cell phone use
- ☐ poor sleep quality
- ☐ the age of a student
- ☒ addiction to smartphones

is

- ☒ correlated
- ☐ not correlated

with

- ☐ cell phone use.
- ☒ poor sleep quality.
- ☐ the age of a student.
- ☐ addiction to smartphones.

b. Complete the statement.

The researchers

- ☒ can
- ☐ cannot

claim that

- ☐ cell phone use
- ☐ poor sleep quality
- ☐ the age of a student
- ☒ addiction to smartphones

causes

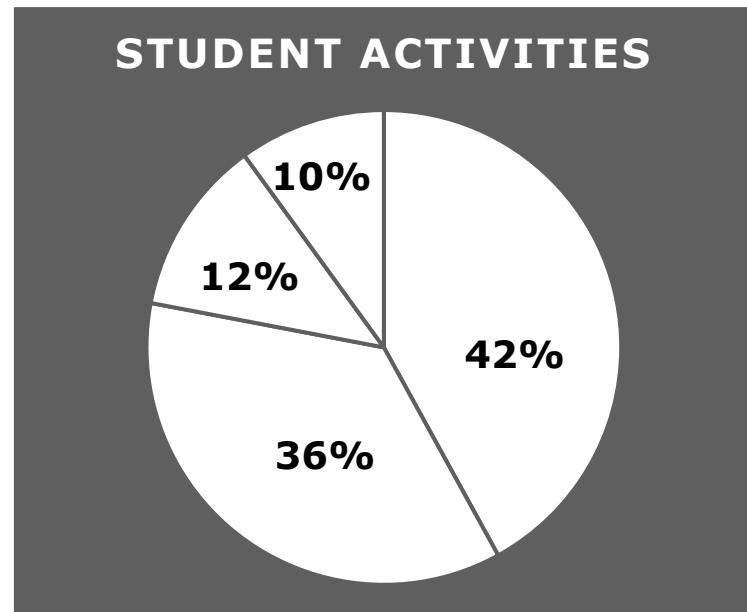
- ☐ cell phone use
- ☒ poor sleep quality
- ☐ the age of a student
- ☐ addiction to smartphones

because

- ☒ the study was a controlled study.
- ☐ correlation does not imply causation.

6. A survey asked 2,450 students which types of activities they are likely to participate in. The results of the survey are shown.

Category	Percentage of Students
Individual or team sports	42%
Art, music, or dance	36%
Computer programing or video games	12%
Debate team or chess club	10%



- a. Determine the number of respondents who are likely to participate in art, music, or dance.

Number of Respondents	882
-----------------------	-----

The original results of the survey did not group the categories. The original results are shown.

Category	Number of Students	Category	Number of Students
Individual sports	637	Team sports	392
Art	196	Music	490
Dance	196	Computer programing	49
Video games	245	Debate team	147
Chess club	98		

- b. Complete the statement.

Based on original survey results the circle graph would have

- ☐ 3
- ☐ 4
- ☐ 5
- ☒ 6
- ☐ 7

sectors

that are labeled as 10% or less and

- ☒ 3
- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 7

labeled as more than 10%.